Ensembling Descendant Term Classifiers to Improve Gene - Abnormal Phenotype Predictions

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Abstract. The Human Phenotype Ontology (HPO) provides a standard categorization of the phenotypic abnormalities encountered in human diseases and of the semantic relationship between them. Quite surprisingly the problem of the automated prediction of the association between genes and abnormal human phenotypes has been widely overlooked, even if this issue represents an important step toward the characterization of gene-disease associations, especially when no or very limited knowledge is available about the genetic etiology of the disease under study. We present a novel ensemble method able to capture the hierarchical relationships between HPO terms, and able to improve existing hierarchical ensemble algorithms by explicitly considering the predictions of the descendant terms of the ontology. In this way the algorithm exploits the information embedded in the most specific ontology terms that closely characterize the phenotypic information associated with each human gene. Genome-wide results obtained by integrating multiple sources of information show the effectiveness of the proposed approach.

Keywords: Human Phenotype Ontology · Hierarchical multi-label classification · Hierarchical ensemble methods · Gene-abnormal phenotype prediction

1 Background

The Human Phenotype Ontology (HPO) project [9] aims at providing a standard categorization of the abnormalities associated with human diseases and the semantic relationships between them. Each HPO term does not represent

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a disease, but rather it denotes individual signs or symptoms or other clinical abnormalities that characterize a disease. The HPO contains approximately 11,000 terms (still growing) and over 115,000 annotations to hereditary diseases. Moreover the HPO provides a large set of HPO annotations to approximately 4000 common diseases. The HPO is structured as a direct acyclic graph (*DAG*), where more general terms are found on the top levels of the hierarchy and the term specificity increases moving from the root to the leaves. Figure 1 shows an example of a small subset of the HPO, including all the HPO nodes that are ancestors of the *Tryptophanuria* term. In this example *Tryptophanuria* is the most specific HPO term, its parent term *Aminoaciduria* is less specific, and following the path toward the root term we find more general terms, such as *Abnormality of the urinary system*, till to the root term *Phenotypic abnormality*.

Each HPO term belongs to one of the following five subontologies: *Phenotypic abnormality*, *Clinical modifier*, *Mortality/Aging*, *Mode of inheritance* or *Frequency*. All the HPO relationships are *is-a* (class-subclass relationships) and are governed by the *true-path-rule* (also known as *annotation propagation rule*) [2] that can be summarized as follow: an annotation for a functional term is

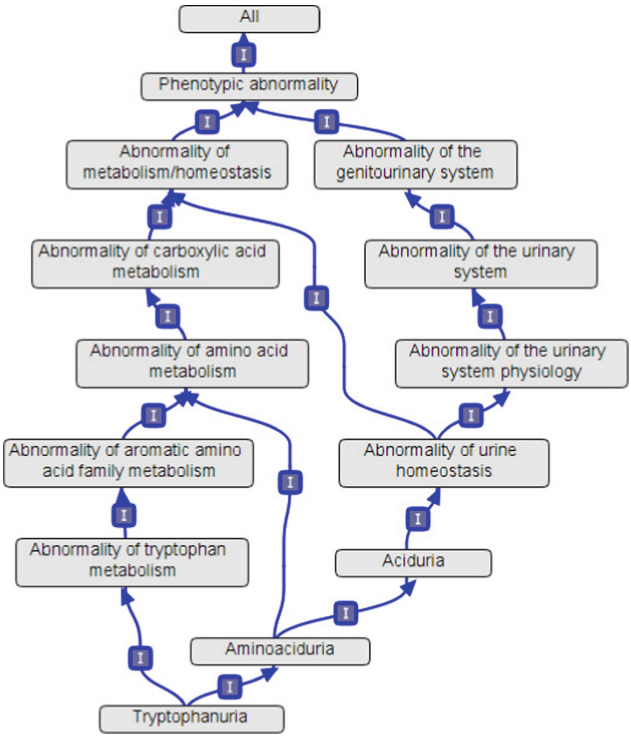

Fig. 1. Ancestor view of the HPO terms Tryptophanuria (*Phenotypic abnormality* subontology). Figure created by using OBO-Edit, an open source ontology editor.

transferred in a recursive way to its ancestors, whereas if a gene is unannotated for a class, it cannot be annotated with its descendants.

While the problem of the prediction of gene–disease associations has been widely investigated [10], the related problem of gene–phenotypic feature (i.e. HPO term) association has been largely overlooked, despite the quickly growing application of the HPO to relevant medical problems [16,22]. In principle in the contest of gene–abnormal phenotype prediction, any "flat" method that predicts labels independently of each other can be applied [21], but it may introduce significant inconsistencies in the classification due to the violation of the *true path rule* that governs the HPO taxonomy. Besides inconsistency, flat methods may also loose important a priori knowledge about the constraints of the hierarchical labeling that could enhance the accuracy of the predictions.

To overcome these limitations we recently proposed an ensemble method (*Hierarchical True path Rule for Directed Acyclic Graph* - TPR-DAG) [11, 13] that explicitly takes into account the hierarchical relationships between HPO terms, and in [11] we showed that TPR-DAG achieves competitive results with respect to state-of-the-art methods for HPO term prediction. More in general ensemble methods have been successfully applied to several branches of bioinformatics, ranging from genetic associations studies to pathogenic genetic variant prediction [7,15]. In this paper we propose a variant of the TPR-DAG algorithm, that we named *DEScendant Classifier ENSemble* (DESCENS). The novelty of DESCENS with respect to TPR-DAG consists in strongly considering the contribution of all the descendants of each node instead of only that of its children, since with the TPR-DAG algorithm the contribution of the descendants of a given node decays exponentially with their distance from the node itself, thus reducing the impact of the predictions made at the most specific levels of the ontology [17]. On the contrary DESCENS predictions are more influenced by the information embedded in the most specific terms of the taxonomy (e.g. leaf nodes), thus putting more emphasis on the terms that most characterize the gene under study, and that are those usually most informative and meaningful from a bio-medical standpoint.

2 Materials and Methods

Let $G =< V, E >$ a Directed Acyclic Graph (DAG) with vertices $V = \{1, 2, \ldots, |V|\}$ and edges $e = (i, j) \in E, i, j \in V$. G represents the HPO taxonomy structured as a DAG, whose nodes $i \in V$ represent classes (terms) of the ontology and a directed edge $(i, j) \in E$ the hierarchical relationships between i (parent term) and j (child term). A "continuous flat multi-label scoring" predictor $f : X \rightarrow [0, 1]$ provides a score $\hat{y}_i \in [0, 1]$ that can be interpreted as the likelihood or probability that a given gene belongs to a given node/HPO term $i \in V$ of the DAG G. The set of $|V|$ flat classifiers provides a multi-label score $\hat{\boldsymbol{y}} \in [0, 1]^{|V|}$: $\hat{\boldsymbol{y}} =< \hat{y}_1, \hat{y}_2, \ldots, \hat{y}_{|V|} >$. We say that a multi-label scoring $\boldsymbol{y}$ is consistent if it obeys the *true path rule*:

$$\boldsymbol{y} \text{ is consistent } \iff \forall i \in V, j \in parents(i) \Rightarrow y_j \geq y_i \tag{1}$$

According to this rule the score of a parent or an ancestor node must be larger or equal than that of its children or descendants nodes.

To process and provide flat scores of the proposed hierarchical ensemble methods we used both a semi-supervised network-based approach (*RANKS* [18]) and a supervised machine learning method (Support Vector Machine – *SVM*). In our experiments we applied *RANKS* with the *average score function* and the *random walk kernel* at 1, 2 and 3 steps, i.e. kernels able to evaluate the direct neighbors and those far away 2 and 3 steps from each gene in the network. It is worth noting that *RANKS* returns a score and not a probability [12]. To make the scores comparable across classes we normalized the scores in the sense of the maximum (i.e. we divided the score values of each class by the maximum score of that class) or according to the quantile normalization [3].

After the learning phase the "flat" predictions are modified by the DESCENS algorithm, whose high-level pseudo-code is shown in Fig. 2.

```
Input:
- G =< V, E >
- V = {1, 2, ..., |V|}
- ŷ =< ŷ_1, ŷ_2, ..., ŷ_|V| >,    ŷ_i ∈ [0, 1]
begin algorithm
01:     A. dist := ∀i ∈ V ComputeMaxDist (G, root(G))
02:     B. Per-level bottom-up visit of G:
03:         for each d from max(dist) to 0 do
04:             N_d := {i | dist_i = d}
05:             for each i ∈ N_d do
06:                 Δ_i := {j ∈ desc(i) | ȳ_j > ŷ_i}
07:                 ȳ_i := 1/(1+|Δ_i|) (ŷ_i + Σ_{j∈Δ_i} ȳ_j)
08:             end for
09:         end for
10:     C. Per-level top-down visit of G:
11:         ŷ := ȳ
12:         for each d from 1 to max(dist) do
13:             N_d := {i | dist_i = d}
14:             for each i ∈ N_d do
15:                 x := min_{j∈parents(i)} ȳ_j
16:                 if (x < ŷ_i)
17:                     ȳ_i := x
18:                 else
19:                     ȳ_i := ŷ_i
20:             end for
21:         end for
end algorithm
Output:
- ȳ =< ȳ_1, ȳ_2, ..., ȳ_|V| >
```

Fig. 2. DEScendant Classifier ENSemble for DAGs (DESCENS)

The block A of the algorithm (row 1) computes the maximum distance of each node from the root. To this end a method based on the Topological Sorting algorithm can be applied [5]. The block B computes a per-level bottom-up visit of the graph G (rows 2 to 9) to propagate the "positive" predictions across the hierarchy. More precisely, according to the true path rule, only the "positive" descendants of a certain node i (e.g. descendant nodes having scores larger than that of their ancestor node i) influence the prediction for the node i itself (row 6 of Fig. 2). In this way all the "positive" descendants of node i provide the same contribution to the ensemble prediction $\bar{y}_i$, by modifying the flat predictions $\hat{y}_i$.

1. *Threshold Free (TF) Strategy.* We choose as "positive" descendants those nodes that achieve a score higher than that of their ancestor node i:

$$\Delta_i := \{j \in desc(i) | \bar{y}_j > \hat{y}_i\} \tag{2}$$

   This strategy leads to the DESCENS-TF algorithm (Fig. 2).
2. *Adaptive Threshold (T) Strategy.* The threshold is selected to maximize some performance metric $\mathcal{M}(j,t)$ (e.g. F-score or *AUPRC*) estimated on the training data for the class j with respect to the threshold t. The corresponding set of positives $\forall i \in V$ is:

$$\Delta_i := \{j \in desc(i) | \bar{y}_j > t_j^*, t_j^* = \arg\max_t \mathcal{M}(j,t)\} \tag{3}$$

   For instance t_j^* can be selected from a set of $t \in (0,1)$ through internal cross-validation techniques. This strategy leads to the DESCENS-T algorithm, simply by changing row 6 in Fig. 2 with Eq. 3.

Moreover, by changing the line 7 of the algorithm in Fig. 2, we can design the "weighted" version of the DESCENS algorithm (DESCENS-W) merely adding a weight $w \in [0,1]$ to balance the contribution between the node i and that of its "positive" descendants:

$$\bar{y}_i := w\hat{y}_i + \frac{(1-w)}{|\Delta_i|} \sum_{j \in \Delta_i} \bar{y}_j \tag{4}$$

Another variant of DESCENS (named DESCENS-τ) balances the contribution between the "positive" children of a node i and that of its "positive" descendants excluding its children by adding a weight $\tau \in [0,1]$:

$$\bar{y}_i := \frac{\tau}{1+|\phi_i|}(\hat{y}_i + \sum_{j \in \phi_i} \bar{y}_j) + \frac{1-\tau}{1+|\delta_i|}(\hat{y}_i + \sum_{j \in \delta_i} \bar{y}_j) \tag{5}$$

where ϕ_i are the "positive" children of i and $\delta_i = \Delta_i \setminus \phi_i$ the descendants of i without its children. If $\tau = 1$ we consider only the contribution of the "positive" children of i, and if $\tau = 0$ only the descendants that are not children contribute to the score, while for intermediate values of τ we can balance the contribution of ϕ_i and δ_i positive nodes.

Independently of which variants of the DESCENS algorithm we decide to use, "positive" predictions are "bottom-up" recursively propagated from the parents towards the ancestors of each node. The bottom-up step does not assure the consistency of the predictions. Therefore, this is guaranteed by the block C of the algorithm (row 10 to 21), where the nodes are top-down processed by level in an increasing order (from the least to the most specific terms) and the "bottom-up" scores computed at the block B are hierarchically corrected to $\bar{y}$ according to the following simple rule:

$$\bar{y}_i := \begin{cases} \hat{y}_i & \text{if} \quad i \in root(G) \\ \min_{j \in parents(i)} \bar{y}_j & \text{if} \quad \min_{j \in parents(i)} \bar{y}_j < \hat{y}_i \\ \hat{y}_i & \text{otherwise} \end{cases} \tag{6}$$

The aim of the top-down step consists in propagating the "negative" predictions towards the children and in a recursive way towards the descendants of each node. Considering the sparseness of the HPO, it is easy to see that the overall computational complexity of DESCENS algorithm is $\mathcal{O}(|V|)$.

3 Results

We downloaded physical and genetic experimental interactions relative to 4970 proteins from BioGRID (v. 3.2.106, [4]) and the integrated protein-protein interaction and functional association data for 18,172 human proteins from STRING (v. 9.1, [6]). Moreover, starting from the Gene Ontology annotations of the three main sub-ontologies (Biological Process, Molecular Function and Cellular Component) and from OMIM annotations [1], both represented as binary feature vectors, we constructed 4 more networks by using the classical Jaccard index to represent the edge weight (functional similarity) between the nodes (genes) of the resulting network. In our context the Jaccard index of two genes measures the ratio between the cardinality of their common annotations and the cardinality of the union of their annotations. The rationale behind the usage of this index is that two genes are similar if they share most of their annotations. All these annotations were obtained by parsing the raw text annotation files made available by Uniprot knowledge-base considering only its SWISSPROT component. Finally the resulting $n = 6$ networks have been integrated by averaging the edge weights w_{ij}^d between the genes i and j of each network $d \in \{1, n\}$ after normalizing their weights in the same range of values $w_{ij}^d \in [0, 1]$ (*Unweighted Average* (UA) network integration, [20]):

$$\bar{w}_{ij} = \frac{1}{n} \sum_{d=1}^{n} w_{ij}^d \tag{7}$$

The resulting weighted adjacency matrix representing the obtained networks is made up of 19,430 human proteins. From the HPO website we downloaded the January 2014 release, by considering the *Phenotypic Abnormality* subontology,

that is the main subontology of the HPO (the other subontologies are significantly smaller and amount to only some tens of terms). To avoid prediction of HPO terms having too few annotations, for a reliable assessment we pruned HPO terms having less than 10 annotations obtaining a final HPO-*DAG* composed by 2154 HPO terms and 2641 between-terms-relationship.

The generalization performance of the methods were assessed through a classical 5-fold cross-validation procedure, whereas the results were evaluated by using the *gene-centric* metric F_{max} (i.e. the maximum hierarchical F-score achievable by "a posteriori" setting an optimal decision threshold [8]) and two *term-centric* metrics: the classical Area Under the Receiver Operating Characteristic Curve (*AUROC*) and the Area Under the Precision Recall Curve (*AUPRC*) to take into account the imbalance of annotated vs. unannotated HPO terms.

Table 1 summarizes the results achieved by the hierarchical methods HTD-DAG [19] and TPR-DAG [11] and by DESCENS, the novel ensemble variant presented in this manuscript.

Table 1. Average *AUROC* and *AUPRC* across terms and average F_{max}, Precision and Recall across genes of HTD-DAG, TPR-DAG and DESCENS ensemble variants using both *RANKS* and *SVMs* as base learner. Results of "flat" *RANKS* and *SVMs* are also reported. Results are estimated through 5-fold cross-validation. Separately for each metric and base learner the results significantly better than the others according to the Wilcoxon Rank Sum Test ($\alpha = 10^{-9}$) are highlighted in bold.

Method	AUROC	AUPRC	F_{max}	Precision	Recall
RANKS (flat)	0.8493	0.0910	0.3106	0.2407	0.4377
HTD-RANKS	0.8506	0.1065	0.3411	0.2717	0.4583
TPR-TF-RANKS	**0.8567**	0.1166	0.3547	0.2880	**0.4615**
TPR-T-RANKS	0.8512	0.1338	0.3574	0.2929	0.4582
TPR-W-RANKS	0.8507	0.1264	0.3620	0.3025	0.4506
DESCENS-TF-RANKS	0.8554	0.1082	0.3679	0.3148	0.4426
DESCENS-τ-RANKS	0.8530	**0.1360**	0.3622	0.3021	0.4520
DESCENS-T-RANKS	0.8503	0.1087	**0.3771**	**0.3227**	0.4535
DESCENS-W-RANKS	0.8502	0.1223	0.3671	0.3071	0.4561
SVM (flat)	0.7128	0.0429	0.1205	0.1165	0.1247
HTD-SVM	**0.8328**	0.0888	0.2597	0.1898	**0.4112**
TPR-TF-SVM	0.7060	0.0525	0.2034	0.1633	0.2694
TPR-T-SVM	0.8297	**0.1036**	0.2611	0.1939	0.3997
TPR-W-SVM	0.7915	0.0909	0.2187	0.1827	0.2723
DESCENS-TF-SVM	0.7092	0.0561	0.2338	0.1877	0.3100
DESCENS-τ-SVM	0.7182	0.0666	0.2424	0.1927	0.3266
DESCENS-T-SVM	0.7940	0.0514	**0.3102**	**0.2796**	0.3483
DESCENS-W-SVM	0.7724	0.0948	0.2373	0.1815	0.3427

In every experiment the hierarchical ensemble methods are able to improve the results of the flat methods used as base learner both in terms of *AUROC*, *AUPRC* and F_{max}. More in detail looking at the results obtained using *RANKS* as base learner, DESCENS-τ and DESCENS-T achieve better results than all the other compared methods in terms of *AUPRC* and F_{max}, while *TPR-TF* achieves the best results in terms of *AUROC*, but HPO classes are highly imbalanced, and in this setting it is well-known that *AUPRC* is a significantly more reliable metric than *AUROC* [14]. Looking at the results obtained using as base learner the *SVMs*, we can observe that, independently of the ensemble method chosen, we achieve a significant strong improvement with respect to the flat prediction, especially in terms of *AUPRC* and F_{max}. Interestingly enough, considering F_{max}, the only hierarchical metric among those considered, DESCENS achieves significantly better results both if we use *RANKS* or *SVMs* as base learners.

Finally we can observe that the performances of hierarchical ensembles largely depend on those of the flat base learners: for instance *DESCENS-τ-RANKS* achieves a significantly higher precision at all recall levels with respect to *DESCENS-W-SVM*, due to the better performance of the *RANKS* base learner (Fig. 3).

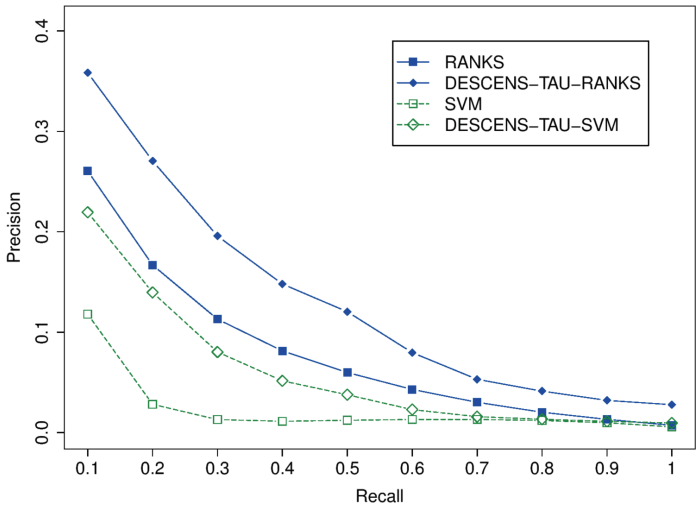

Fig. 3. Compared precision at different recall levels averaged across 2153 HPO terms of *DESCENS-τ* using *RANKS* and *SVM* as base learner. The results of the corresponding flat methods, *RANKS* and *SVM* are also reported.

This is not surprising since the improvement introduced by hierarchical ensemble methods also depends on the predictions of the underlying flat base learner: DESCENS can improve the flat predictions, but there is no guarantee of a correct prediction if most of the base flat learners provide incorrect predictions (Fig. 4).

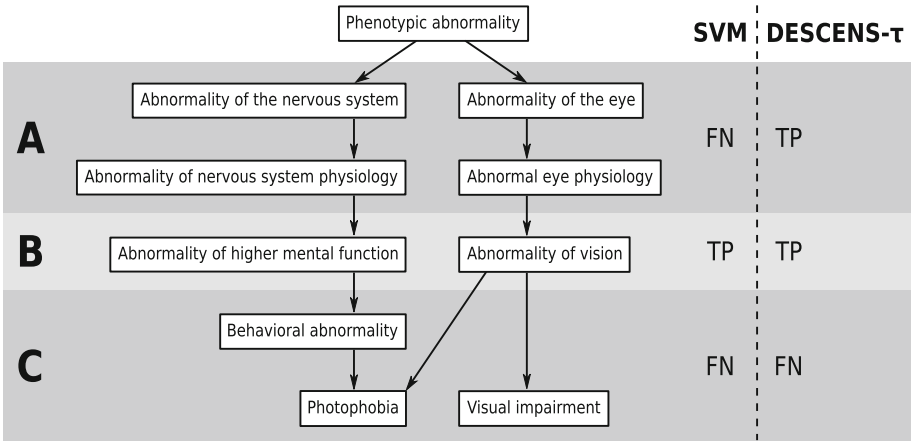

Fig. 4. Flat (*SVM*) and hierarchical *DESCENS-*τ HPO predictions for the gene *RGS9*. At the right side are displayed the correct TP and the incorrect FN predictions made respective by flat-*SVM* and by hierarchical *DESCENS-*τ. In the A box are depicted the predictions that the hierarchical method was able to correct with respect to flat method ($FN \rightarrow TP$); in B are portrayed the correct predictions for both flat and hierarchical methods and finally in C are shown the incorrect flat predictions that the hierarchical method was not able to recover.

4 Conclusion

Genome and ontology wide experimental results show that the DESCENS algorithm is able to improve the predictions of both semi-supervised flat methods, such as the *RANKS* algorithm, that resulted one of the top ranked method in the recent *CAFA2* challenge for HPO term prediction [8], and of supervised methods such as *SVMs*, in terms of *AUROC*, *AUPRC* and F_{max}. Moreover DESCENS further improves HTD-DAG, and TPR-DAG, two of the state-of-the-art methods for HPO prediction, in terms of both *AUPRC* and F_{max}. Furthermore the proposed ensemble methods always provide consistent predictions that obey the *true path rule*, a fundamental fact to assure biologically coherent predictions among HPO terms.

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